CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Nowadays, we see the chatbots everywhere, chatbots are the source of answers to the users questions in any particular domain where it is operating. Chat bots are the source of answers to the users questions in any particular.

The rapid growth of technology has reshaped many sectors, including education, where institutions now use digital tools to enhance learning, streamline administration, and improve communication. One key development is the rise of chatbots AI-powered systems that mimic human conversations to provide automatic responses. Initially rule-based, chatbots have advanced with AI and NLP,becoming more accurate and capable. Research, such as by (Adamopoulou & Moussiades, 2020), shows chatbots are effective across various fields, including education, where they enhance accessibility and efficiency.

NLP, is also an AI technique that enables chatbots to simply control, understand and interpret users queries in natural languages. It is built from the fields of AI, computer science and linguistics which aim towards achieving easy interactions between computers and users in human languages. Importantly, there is embedded information or large chunk of natural language data being stored in a chatbot database by the programmer to recognize different structures of sentences (Hussain, Folu, and Shrish, 2021).

Studies by Okonkwo et al., (2022) highlight that chatbots in higher education reduce theworkload on administrative staff by answering common questions on admissions, courses, and schedules, leading to greater student satisfaction. At Federal Polytechnic Ede, the Computer Science Department faces challenges with traditional methods of handling high volumes of queries. The delays frustrate users seeking quick information, underscoring the need for a chatbot to handle routine inquiries.

1.2 Statement of the Problem

The Computer Science Department at Federal Polytechnic Ede relies on traditional methods of responding to student and public inquiries, which often causes delays, inefficiencies, and dissatisfaction among users. Students and applicants seeking information on admissions, timetables, departmental activities, and staff contacts experience slow responses limited to office hours, creating a communication gap in an era where quick and reliable access to information is expected.

At the same time, administrative staff face an increasing workload as they repeatedly attend to routine questions that could easily be automated. While studies have shown that chatbots in higher education enhance accessibility, reduce administrative burden, and improve responsiveness (Okonkwo et al., 2021) such innovation has not yet been adopted in the department. This highlights the need for a web-based chatbot capable of providing instant, accurate, and automated responses, thereby improving communication, reducing staff workload, and aligning the department with global trends in digital transformation.

1.3 Aim and Objectives of the Study

The primary aim of this study is to design and implement a web-based chatbot for the Computer Science Department at Federal Polytechnic Ede, Osun State, which will grant users access to accurate and timely information. Specific objectives are to:

1. identify the key information and support services most frequently required by students in Computer Science Department at Federal Polytechnic Ede.
2. design and develop a web-based chatbot capable of delivering instant and automated responses to student queries.
3. implement the chatbot with computer science department information systems to provide real-time and accurate updates.

4. evaluate the effectiveness of the chatbot in improving information access and student support services.

1.4 Scope of the Study

The scope of this study is limited to the development and implementation of a web-based chatbot for the Computer Science Department at Federal Polytechnic, Ede, Osun State. The chatbot is designed to handle queries related to course options and timetables, department and staff contact information, departmental events and news, admission requirements and procedures, as well as general administrative processes.

1.5 Significance of the Study

The significance of this study lies in its contribution to improving communication and information access within the Computer Science Department of Federal Polytechnic, Ede. The developed web-based informational chatbot will enhance information accessibility by providing 24/7 instant responses to students, staff, and prospective applicants, thereby eliminating delays caused by manual inquiry handling. It will also reduce administrative workload by automating the handling of routine queries, allowing departmental staff to focus on more demanding and strategic tasks. Furthermore, the system will improve user experience by delivering quick, accurate, and user-friendly interactions, which will increase satisfaction among users. In addition, the chatbot will promote efficient communication by ensuring timely dissemination of important departmental updates, announcements, and academic information, ultimately supporting effective information management within the department.

1.6 Choice of Programming Language

In designing and implementing the Web-Based Informational Chatbot for the Computer Science Department at Federal Polytechnic Ede, the use of the following programming languages is considered most appropriate:

1. Python
2. JavaScript
3. PHP
4. MySQL
5. HTML & CSS

Python is ideal because of its simplicity and powerful libraries for natural language processing and machine learning, which makes the chatbot intelligent and adaptable. **JavaScript** is suitable for creating an interactive and real-time user interface that works seamlessly on both desktop and mobile browsers. **PHP** is appropriate for server-side scripting as it is affordable, secure, and integrates easily with **MySQL**, which is useful in the Nigerian educational context. SQL is necessary for efficient storage and retrieval of departmental data and chatbot interactions. **HTML** and **CSS** are essential for structuring and styling the chatbot interface, ensuring it is simple, responsive, and user-friendly.

1.7 Operational Definition of Terms

Artificial Intelligence (AI): Artificial Intelligence refers to the simulation of human intelligence processes by computer systems, particularly in learning, reasoning, and problem-solving tasks, which enables the chatbot to interact intelligently with users.

Chatbot: A chatbot is an artificial intelligence-based software application designed to simulate human conversation and provide automated responses to user queries through a web interface.

Computer Science Department: The Computer Science Department at Federal Polytechnic Ede serves as the specific environment where the chatbot system will be designed, implemented, and evaluated for handling informational queries.

Database: A database is an organized collection of data stored and managed electronically to enable efficient retrieval, updating, and management of information within the chatbot system.

Implementation: Implementation refers to the process of developing, coding, testing, and deploying the web-based chatbot system to ensure it functions as designed in a real-world environment.

Information System: An information system is a structured setup that collects, processes, stores, and distributes data to support decision-making, coordination, and control within an organization.

Knowledge Base: A knowledge base is a structured repository of information and responses that the chatbot uses to answer user questions and provide accurate, relevant information.

User Interface (UI): The user interface is the visual part of the chatbot system through which users interact with the application, typically consisting of input fields, chat windows, and response displays.

Web Server: A web server is a computer system that hosts the chatbot application, processes requests from users, and delivers responses through a web browser.

Web-Based Application: A web-based application is a software system that operates on a web server and can be accessed through an internet browser without requiring installation on a local computer.

CHAPTER TWO

LITERATURE REVIEW

The integration of chatbots in education has surged recently, driven by advancements in AI and NLP, making these tools essential for round-the-clock support in administrative tasks, student inquiries, and learning assistance. This chapter reviews existing literature on educational chatbots, covering their applications, benefits, challenges, and trends, to support the design of a chatbot for the Computer Science Department at Federal Polytechnic Ede.

2.1 Evolution of Chatbot Technology in Education

Chatbots have evolved significantly since the 1960s, with early systems like Joseph Weizenbaum's ELIZA offering basic pattern-matching responses. Modern chatbots, powered by AI advances and deep learning techniques such as RNNs and transformers, now provide sophisticated, context-aware interactions. They offer personalized responses, assist in problemsolving, and integrate with educational management systems, delivering real-time, up-to-date information to students and staff (Smutny & Schreiberova, 2020; Winkler & Söllner, 2022).

Natural language understanding (NLU) and generation (NLG) advancements have also enhanced chatbots' ability to accurately interpret complex queries and provide coherent, contextually relevant responses. This evolution has positioned chatbots as valuable tools for improving engagement, operational efficiency, and personalized support in educational settings.

Over the last half-century, the exponential progression of computer technology has profoundly impacted every facet of our daily lives. Initially exclusive to workplaces as a computational device, computers have embarked on a transformative journey, metamorphosing through several evolutionary phases. From being a mere tool for data processing, they have been all over our households in the form of personal computers.

As we stand today, this technology has morphed into a ubiquitous, hand-held accessory, which the majority of individuals carry within their pockets. Functioning as an intuitive assistant, these devices maintain perpetual connectivity to the internet, equipping us to perform a myriad of tasks that cater to a diverse spectrum of user requirements. This transformation illustrates the capacity of technology to adapt and evolve, mirroring the multifaceted needs of its user base.

2.2 Multi-Domain Impact of Computer Technology: A Review Across Diverse Sectors

The ubiquity of computers in our society has been extensively documented in scholarly research across various disciplines. They have become integral tools, utilized in diverse sectors ranging from healthcare to finance, from entertainment to education, each leveraging its unique capabilities to drive efficiency, innovation, and accessibility.

In the realm of healthcare, computers have revolutionized the delivery and management of care. A study by (Hillestad et al., 2025) highlights how health information technology can potentially save over $81 billion annually in the United States by enhancing efficiency and

safety. Similarly, advancements like telemedicine and electronic health records have empowered remote healthcare delivery and data-driven decisions (Mair & Whitten, 2020).

The finance sector has seen transformational changes through computer-based algorithms, enabling highfrequency trading and complex financial modeling (Aldridge, 2023). In addition, financial technologies, or 'fintech', have broadened access to financial services through digital platforms (Zavolokina et al., 2023).

Lastly, the entertainment industry has experienced remarkable shifts due to computer technology. Burgess & Green, (2022) examined how platforms like YouTube have changed media production and consumption, offering more people the chance to create and share media. Alongside this, (Manovich, 2021) explored the wider impact of digitization, which has not just changed how we create and access media, but also reshaped our ideas of 'media', 'computation', and 'data'.

These examples highlight the broad and deep influence of computer technology across various sectors. As technology continues to evolve, it is poised to affect more areas of society, constantly responding to and shaping our changing needs. The implications of this in education will be discussed in detail in the following sections.

2.3 Artificial Intelligence in Education

Artificial Intelligence (AI) has made significant steps within the field of education, offering innovative methods to enhance teaching and learning. This section will explore several recent studies that illustrate these advancements.

Several studies have explored the application of AI chatbots in education, shedding light on their potential as intelligent student assistants. One such study by (Pereira, 2021) titled "Leveraging chatbots to improve self-guided learning through conversational quizzes" investigated the use of chatbots as a tool to enhance self-guided learning. The study developed a chatbot called Dawebot, implemented on the Telegram platform, which

facilitated interactive quizzes and provided immediate feedback to students. The study involved 20 participants, and the chatbot recorded their quiz attempts and progress. The results showed that students found the chatbot experience valuable, and the majority agreed that using bots for practicing tests was a beneficial idea. The chatbot offered features like immediate feedback, ease of use, and continuous progress tracking. The findings suggest that chatbots have the potential to be an effective tool for selfguided learning, offering personalized learning experiences and aiding students' engagement in their subjects.

Similarly, Ho (2020) conducted a study titled "TA-bot: An AI agent as a Teaching Assistant" that presented the development and implementation of an artificial intelligence agent designed as a virtual teaching assistant. The agent, TA-bot, was created by utilizing existing conversational technologies and commercial off-the-shelf software. The study addresses the ongoing problem of staff and resource shortages in educational settings by proposing the use of AI teaching assistants to augment the existing teaching staff. The study demonstrates the successful utilization of TA-bot in a Digital Marketing class at George Mason University, where the agent serves as a specialized assistant, providing information and answering student inquiries related to the course syllabus. The study concludes that virtual teaching assistants, such as TAbot, can be effectively created using readily available technologies. The results indicate that current natural language processing capabilities enable efficient training and deployment of such assistants, thereby enhancing the learning experience for students.

Another innovative architecture called Rexy, designed for constructing virtual teaching assistants (VTAs), was introduced in a study by (Benedetto & Cremonesi, 2022). The study introduces an innovative architecture called Rexy, designed for constructing virtual teaching assistants (VTAs). The application utilizes IBM's Watson Assistant to facilitate natural language understanding and processing tasks. The study presents the implementation of Rexy in an on-site course at Politecnico di Milano and evaluates its effectiveness through a

qualitative user study. The results indicate positive feedback from students, highlighting the VTA's ability to provide immediate answers and reduce the workload on professors. The study also identifies areas for improvement, such as enlarging the training set to enhance intent recognition and optimizing the confidence threshold for generating answers. Overall, the study concludes that Rexy offers a promising solution for implementing VTAs and suggests future work to further refine its performance and expand its capabilities, including exploring one-to-many interactions.

The use of chatbots as interactive tools for language education was explored in a study by Dokukina and Gumanova (2020) titled "The rise of chatbots – new personal assistants in foreign language learning". The authors discuss the development of chatbot technologies, starting from early attempts in the 1960s to the more sophisticated and intelligent chatbots of today, such as Microsoft's XiaoIce. The study highlights the role of chatbots in language learning, emphasizing their ability to provide structured interactions, mentoring, and practice for specific language skills. Examples like Vasya and Duolingo are presented as effective chatbot-based language tutorials. The conclusion of the study suggests that chatbots offer convenient, accessible, and interactive learning experiences for language learners, particularly in the fastpaced modern world. While chatbots cannot replace human teachers, they can serve as valuable tools to facilitate and enhance language learning processes, catering to the needs and preferences of individual learners.

Examining the evolution of chatbot technology (Dimitriadis, 2020) conducted a study titled "Evolution in Education: Chatbots" that explored the potential application of chatbots in educational settings. The study highlights how chatbots, which are AI programs simulating interactive human conversation, have become a prominent trend in the global market. It discusses the advantages and disadvantages of using chatbots in various sectors and emphasizes their potential for personalized and adaptive learning in education. Furthermore,

the study investigates the possibility of chatbots serving as virtual teaching assistants, relieving teachers from repetitive tasks and allowing them to focus more on providing quality education. It delves into the psychological dimensions of human interaction with chatbots, emphasizing the tendency to anthropomorphize these virtual assistants. The study concludes that while chatbots offer significant benefits, their integration into education is still in its early stages, and further research is needed to fully harness their potential in enhancing the learning process.

A study by Chen et al., (2022) titled "Artificial Intelligence (AI) Student Assistants in the Classroom: Designing Chatbots to Support Student Success" investigated the role of chatbots in higher education. The research aimed to examine the effectiveness of chatbots as interactive and responsive learning tools, focusing on their ability to teach basic AI concepts, enhance student engagement, and alleviate the workload of educators. The study found that students generally perceived chatbots as valuable additions to the classroom, appreciating their interactive nature and responsiveness. However, the research also highlighted certain limitations of chatbots, notably their inability to fully replicate human conversation or emotional responses. Recognizing the need for improvement, the study called for advancements in chatbots' intelligence and emotional understanding, along with a heightened consideration of ethical implications surrounding their usage in educational settings. By adding these findings to the existing body of literature, this study contributes to the ongoing exploration of AI chatbots in education.

The integration of AI methods in the education sector was explored in a study by Haderer & Ciolacu, (2022) titled "Education 4.0: Artificial Intelligence Assisted Task- and Time Planning System". The study aims to enhance students' self-regulated learning. The paper presents an AI-based system that focuses on task and time planning for students, aiming to simplify their daily routines and support their organization and productivity. The study

emphasizes the importance of digital tools and AI in Education 4.0, highlighting technologies such as adaptivity, mobile connectivity, personalization, AI advice and support, gamification, learning analytics, e-assessments, and the Internet of Things (IoT). The proposed system utilizes AI engines to generate and manage tasks and appointments for students, providing personalized recommendations and adaptive learning experiences. The architecture of the system includes an information base, AI engines, an open API service, a user interface, and an AI dashboard. The study emphasizes the need for data integrity, data protection, and data security while ensuring user-friendliness and usability. The conclusion suggests that the integration of AI and digital tools in education can significantly improve the teaching and learning experience, leading to enhanced efficiency and effectiveness in Education 4.0.

Lastly, Boateng et al., (2022) conducted a study titled "Kwame for Science: An AI Teaching Assistant for Science Education in West Africa" to address educational challenges faced by students in Africa. Particularly the limited access to teachers and difficulty in obtaining answers to their questions. The researchers extended their previous work on Kwame, an AI teaching assistant for coding, to adapt it for science education. Kwame for Science utilized a Sentence-BERT-based question-answering approach and was specifically designed to answer science questions based on the Integrated Science subject of the West African Senior Secondary Certificate Examination (WASSCE). Through a real-world deployment as a web app, the study evaluated the system's performance and showed promising results, with a top 3 accuracy of 87.5% across 190 users from 11 countries. The conclusion of the study highlighted the potential of Kwame for Science in delivering scalable, cost-effective, and quality remote education to millions of students across Africa, with future plans to improve the system's accuracy and expand its availability in local languages and offline channels.

These studies collectively contribute to the exploration of AI in education, demonstrating the potential of chatbots and AI-based systems to enhance learning experiences, provide personalized assistance, and address educational challenges.

2.4 Review of Related Works

The adoption of chatbots in education has gained considerable momentum, with numerous studies highlighting their transformative role in improving administrative efficiency and enhancing student engagement. This section provides an indepth review of recent research projects and implementations of chatbots in educational settings, offering insights into their effectiveness, challenges, and potential for future development.

Winkler & Söllne,r (2022) in their study presented at the Australasian Conference on Information Systems, examined the potential of chatbots to improve student success in higher education. They proposed a chatbot designed to provide personalized learning support and found that such systems could significantly enhance the learning experience by offering immediate, tailored assistance to students. The authors emphasized the importance of integrating chatbots with existing learning management systems to provide a seamless experience for users.

Building on this work, (Smutny & Schreiberova, 2020) conducted a comprehensive review published in Computers & Education, analyzing chatbots used for educational purposes across various fields. Their findings indicated that chatbots were particularly effective in language learning, programming education, and providing administrative support. The authors emphasized the potential of chatbots to offer personalized learning experiences and reduce educators' workload. They also highlighted the need for more sophisticated natural language processing capabilities to handle complex student queries effectively.

In a related study, Pérez et al., (2020) investigated the use of chatbots in higher education for improving student engagement. Their work, published in the International Journal of

Interactive Multimedia and Artificial Intelligence, demonstrated how chatbots could effectively handle student queries about course content, schedules, and administrative procedures, leading to increased student satisfaction and reduced workload for staff. The researchers noted that chatbots were particularly effective in providing instant responses to common questions, allowing human staff to focus on more complex issues that required personal attention.

Adamopoulou & Moussiades, (2020), in their work Chatbots: Hisory, Technology, and Applications, analyzed the evolution of chatbots and their applications in various industries, including education. The authors discussed how advances in natural language processing (NLP) have enabled chatbots to better understand user queries and deliver more accurate responses. Their research highlights the technological advancements that have made chatbots an efficient solution for automating student services and improving information accessibility.

Hien et al., (2022), in their study published in the Journal of Telecommunications and Information Technology, explored the use of chatbots in university information systems. They proposed a model for integrating chatbots with existing university databases to provide students with quick access to information about courses, schedules, and academic procedures. The researchers emphasized the importance of data security and privacy when implementing such systems, particularly given the sensitive nature of student information.

Extending the scope of chatbot applications, Ondáš et al., (2022) published a study in the IEEE Access journal, focusing on the use of conversational AI in e-learning environments. Their work demonstrated how chatbots could be used to create interactive learning experiences, particularly in the context of language education. The authors highlighted the potential of voice-enabled chatbots to improve students' speaking and listening skills, suggesting a new frontier in educational technology.

In a similar vein, Okonkwo and Ade-Ibijola (2021) presented their findings in the International Journal of Artificial Intelligence in Education, showcasing a chatbot designed to assist computer science students with programming concepts. Their system used natural language processing to interpret student queries and provide explanations, code examples, and debugging assistance. The authors noted significant improvements in student understanding and engagement, particularly for complex programming topics.

Addressing the challenge of scalability in educational support, Cunningham-Nelson et al. (2022) published their work in the Australasian Journal of Educational Technology. They developed a chatbot capable of handling a high volume of student queries during peak periods, such as exam times or course registration. The system demonstrated the ability to significantly reduce response times and improve overall student satisfaction with support services.

Focusing on the personalization aspect of educational chatbots, (Hobert, 2022) presented a study in the Proceedings of the 52nd Hawaii International Conference on System Sciences. The author explored the use of adaptive learning techniques in chatbot design, allowing the system to tailor its responses based on individual student profiles and learning progress. This approach showed promise in providing more effective and engaging learning experiences for students with diverse needs and learning styles.

Lastly, Molnár & Szüts, (2022) published their findings in the Journal of Applied Technical and Educational Sciences, examining the ethical considerations of using AI-powered chatbots in educational settings. Their work highlighted the importance of transparency in chatbot interactions, ensuring that students are aware when they are communicating with an AI system. The authors also discussed the potential biases that could be introduced through chatbot algorithms and the need for careful design and monitoring to ensure equitable support for all students.

2.5 Research Gap

Despite the growing body of research on the use of conversational AI chatbots in education, many studies to date have focused primarily on the chatbots' abilities to answer questions based on specific curriculums without the provision of associated textbooks (Chen et al., 2022; Pereira, 2021; Benedetto & Cremonesi, 2022). These models are typically pre-trained on specific educational materials and are thus limited in their flexibility and adaptability to cater to different learning materials and diverse educational needs.

One noteworthy exception is the study by Boateng et al., (2022), which proposed a system that allowed users to provide their own textbooks for the AI to generate responses. Nonetheless, the system required further training on the provided materials to deliver accurate responses, a time-consuming process that could potentially hinder its wide-scale application.

Despite its pioneering efforts, Boateng et al., (2022) study illuminates a significant research gap: the lack of chatbot applications capable of generating accurate responses from user-provided textbooks without the need for additional training. It also underscores the need for the development of more flexible, adaptable, and user-friendly educational chatbots.

The present study aims to address this research gap by using the GPT, a pre-trained Large Language Model (LLM), in conjunction with the llama-index library, which has the capacity to read and index documents without the need for additional training. This approach not only eliminates the time and resources required for training but also enhances the flexibility and adaptability of chatbots to cater to a wider range of educational materials and user requirements.

CHAPTER THREE

SYSTEM METHODOLOGY AND DESIGN

3.1 Introduction

This chapter presents the methodological and design framework adopted for the development of the web-based informational chatbot system. It outlines the system development methodology, system analysis, design approach, system architecture, and database structure. The essence of this chapter is to demonstrate the systematic process through which the chatbot was conceptualized, developed, and prepared for implementation.

The goal of the system is to create an intelligent, user-friendly, and accessible web-based platform that provides instant responses to departmental inquiries, thereby improving communication efficiency within the Department of Computer Science, Federal Polytechnic, Ede.

3.2 System Development Methodology

A System Development Life Cycle (SDLC) approach was adopted to guide the process of analysis, design, development, testing, and deployment. Among the various SDLC models, the Iterative Waterfall Model was chosen for this project because of its structured sequence of stages, clarity of documentation, and allowance for feedback loops during development.

The stages include:

1. Problem Identification: Recognizing the limitations in the existing manual information dissemination method.

2. Feasibility Study: Assessing the practicality and technical possibility of developing a chatbot to address the communication gap.
3. System Analysis: Understanding the current workflow, identifying user requirements, and defining system functionalities.
4. System Design: Developing system architecture, data flow diagrams, and user interface models.
5. Implementation: Writing the code and integrating the designed modules.
6. Testing and Evaluation: Ensuring the system performs as expected and meets all functional requirements.
7. Maintenance: Regular updates, debugging, and expansion of the chatbot's knowledge base.

3.3 System Analysis

System analysis involves the critical study of the existing information dissemination process and the specification of requirements for the proposed solution.

3.3.1 Description of the Existing System

Currently, students and staff in the Department of Computer Science depend largely on traditional means of asking questions or querry, such as notice boards, memos, and verbal announcements for accessing departmental information. This traditional system is inefficient, time-consuming, and often unreliable due to delays or missed communications.

3.3.2 The Proposed System

The proposed web-based informational chatbot system aims to solve the above challenges by providing an automated digital platform capable of delivering real-time responses to students' inquiries. The chatbot, powered by a pre-trained knowledge base, will act as a virtual departmental assistant that handles queries such as lecture schedules, course registration details, and staff contact information.

This approach enhances efficiency, accessibility, and transparency while reducing the workload on administrative staff.

3.4 System Design Overview

The system design transforms the analyzed requirements into a structured framework that guides the software's development and implementation. It defines how the system components will interact, how data will flow, and how user inputs will be processed.

The Model-View-Controller (MVC) design pattern was adopted for this project to ensure modularity, flexibility, and maintainability of the codebase.

1. Model: Manages the data and logic of the chatbot (database and knowledge base).
2. View: Represents the chat interface displayed to users.
3. Controller: Serves as the intermediary between the user and the system logic.

3.5 System Architecture

The architecture of the system follows a three-tier web-based structure, consisting of:

Presentation Layer (Frontend):

This layer provides the interactive interface through which users (students and staff) communicate with the chatbot. It is implemented using HTML, CSS, and JavaScript, rendered through Python's Flask templates.

Application Layer (Logic Layer):

This layer contains the chatbot's processing logic implemented in Python. It interprets user queries, accesses the knowledge base, and returns intelligent responses. Libraries such as

ChatterBot, NLTK, and Regular Expressions (re) were utilized for natural language processing and text matching.

Database Layer (Backend):

The database layer was implemented using SQLite3, a lightweight yet powerful relational database included by default in Python. It stores chatbot training data, user interactions, and administrative content updates.

3.6 System Design Models

To effectively illustrate the structure and behavior of the system, several models were developed using the Unified Modeling Language (UML).

3.6.1 Use Case Diagram

The Use Case Diagram depicts the interaction between system actors and functions.

Actors:

1. User (Student/Staff)
2. Administrator

Use Cases:

1. Send Message / Ask Question
2. Receive Response
3. View Departmental Information
4. Manage Chatbot Responses (Admin)
5. View Chat Logs (Admin)

Description:

Users interact with the chatbot through a simple web interface, while the administrator maintains and updates the chatbot's data repository.

3.6.2 Activity Diagram

The Activity Diagram represents the flow of activities as users engage with the chatbot:

- User accesses the chatbot webpage.
- User inputs a query.
- The system receives and processes the query.
- The chatbot searches for a matching response.
- The appropriate answer is displayed.
- If no match exists, an alternative suggestion is provided or the user is prompted to rephrase.

3.6.3 Sequence Diagram

The sequence diagram illustrates the chronological interaction between the system entities:

- User Interface sends query →
- Flask Controller interprets input →
- Chatbot Engine (Model) processes and queries SQLite database →
- Database returns matching response →
- Controller sends response to User Interface for display.

3.7 Database Design

The database was designed to store information efficiently, ensuring reliable retrieval and response generation.

3.7.1 Database Description

The database comprises four major tables:

Table Name	Description	Fields

users	Stores information of active sessions	user_id, username, session_id, timestamp
chat_logs	Records all interactions between the chatbot and users	chat_id, user_query, bot_response, time_stamp
knowledge_base	Holds question-response pairs	kb_id, question, answer, category
admin		

3.7.2

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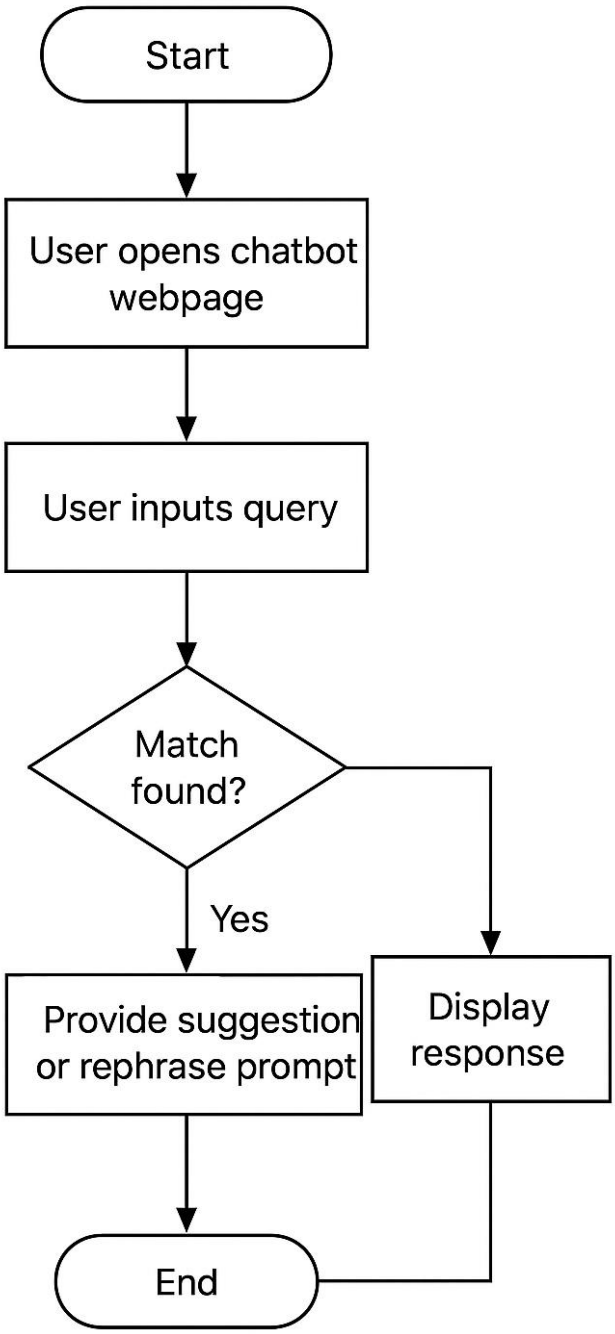
3.8

The sys e chatbot system.



3.9 Tools and Technologies Used

Technology/Library	Purpose
Python 3.12	Core programming language
Flask	Web framework for handling HTTP requests
SQLite3	Local database storage
ChatterBot / NLTK	Chatbot engine and natural language processing
HTML, CSS, JavaScript (Bootstrap)	Frontend development
AJAX / Fetch API	Asynchronous message transmission
Jinja2	Template engine for dynamic webpage rendering

CHAPTER FOUR

SYSTEM DESIGN AND IMPLEMENTATION

4.1 Introduction

This chapter presents the detailed design and implementation of the web-based informational chatbot developed for the School of Computing, Federal Polytechnic Ede. The chapter translates the system requirements and analysis presented in the earlier chapters into a functional software solution. Emphasis is placed on the architectural design, system requirements, chatbot logic, implementation technologies, and operational workflow of the system. According to Pressman and Maxim (2020), effective system design serves as the foundation for successful software implementation by ensuring that functional and non-functional requirements are systematically addressed.

4.2 System Architecture Design

The system is designed using a three-tier web-based architecture, which separates concerns and enhances scalability, maintainability, and security. This architectural model is widely adopted in modern web application development and institutional information systems (Bass et al., 2022).

At the presentation tier, users interact with the chatbot through a web browser interface. This interface provides a conversational window where students and staff of the School of Computing can submit queries and receive responses in real time. The interface is designed to be intuitive and responsive to accommodate users with varying levels of computer literacy.

The application tier houses the chatbot engine and server-side logic. This layer processes user inputs, performs keyword or intent matching, retrieves appropriate responses from the knowledge base, and sends the output back to the user interface. By isolating the chatbot logic at this level, system updates and enhancements can be carried out without affecting the user interface.

The knowledge tier contains the structured repository of institutional information used by the chatbot. Rather than supporting complex transactions, this layer functions primarily as a knowledge base that stores predefined questions, keywords, and corresponding responses related to academic programmes, departmental information, admission guidelines, and school policies.

4.3 System Requirements

System requirements define the conditions under which the chatbot can operate effectively. These requirements are categorized into hardware, software, and user requirements, and they ensure that the system performs optimally in its intended environment.

4.3.1 Hardware Requirements

- Personal computer with a minimum of 4GB RAM

- Functional processor
- Keyboard for user input
- Mouse or touchpad for navigation
- Display screen/monitor
- Server machine capable of running a local or hosted web server

The system is designed to operate on standard computing devices commonly available within the institution. A personal computer with a minimum of 4GB RAM and a functional processor is sufficient for hosting and accessing the system. Since the chatbot is web-based, no specialized hardware is required beyond basic input and output devices such as a keyboard, mouse, and display screen. On the server side, a modest machine capable of running a local or hosted web server is adequate, making the system cost-effective and suitable for an academic environment.

4.3.2 Software Requirements

- Windows or Linux operating system
- Apache web server (via XAMPP or similar environment)
- Server-side scripting language (e.g., PHP)
- Relational database management system or structured data storage (e.g., MySQL or JSON)
- Web browser such as Google Chrome or Mozilla Firefox

The development and deployment of the chatbot require a stable operating system such as Windows or Linux, which supports web server technologies. A web server environment, such as Apache provided through XAMPP, is required to host the backend application and manage HTTP requests. The chatbot logic is implemented using a server-side scripting language, while a relational database management system or structured data format is used to store chatbot knowledge and interaction logs.

Modern web browsers such as Google Chrome or Mozilla Firefox are required on the client side to ensure proper rendering of the chatbot interface and smooth interaction. These software components collectively enable seamless communication between users and the chatbot system.

4.3.3 User Requirements

The primary users of the system are students and staff members of the School of Computing. Users are not required to possess advanced technical skills, as the chatbot interface is designed to support natural language queries. The administrator, however, is expected to have basic computer literacy and minimal technical knowledge to update chatbot responses and manage the knowledge base when institutional information changes.

4.4 System Design Models

4.4.1 Use Case Description

The chatbot system supports two main actors: the user and the administrator. Users interact with the chatbot by entering questions related to academic and administrative information, while the system responds automatically based on stored knowledge. The administrator is responsible for maintaining the relevance and accuracy of the chatbot by updating responses and monitoring system usage. Use case modeling is essential in clarifying system functionality and interaction boundaries (Sommerville, 2016).

4.4.2 System Flow Description

The operational flow of the system begins when a user accesses the chatbot webpage. Upon entering a query, the system preprocesses the input by converting it into a standardized format. The processed query is then compared against the chatbot knowledge base. If a relevant match is found, the corresponding response is returned to the user. In cases where no match exists, the system provides a default response prompting the user to rephrase the

question. This sequential flow reflects standard conversational agent processing models discussed in recent chatbot research (Adamopoulou & Moussiades, 2020).

4.5 Chatbot Design Approach

The chatbot adopts a rule-based and intent-matching design approach, which is particularly suitable for institutional information systems. This approach relies on predefined rules and keyword associations to generate accurate responses. Given that the information required by users such as course details, departmental offices, and academic schedules is largely structured and predictable, a rule-based model ensures reliability and ease of maintenance. Studies have shown that such chatbots are effective in academic environments where response accuracy is critical (Nuruzzaman & Hussain, 2018).

4.6 Knowledge Base Design

The chatbot system does not rely on a traditional transaction-driven database, as it is primarily intended to provide informational support rather than handle dynamic data updates or complex transactions. Instead, the system utilizes a knowledge-based repository, which stores institutional information in a well-structured and easily accessible format. This approach allows the chatbot to deliver accurate and consistent responses to user queries without the overhead of complex database operations.

The knowledge base is structured around mapped relationships between user intents, keywords, and predefined responses. Each potential user query is analyzed for relevant keywords, which are then associated with a corresponding intent or category. For example, a query about admission requirements is linked to the "Admissions" intent, allowing the chatbot to retrieve the appropriate response efficiently. This mapping ensures that even variations in user phrasing can be correctly interpreted, improving the system’s ability to understand and respond to a wide range of questions.

Additionally, the knowledge base includes interaction logs and metadata, which allow administrators to monitor frequently asked questions, identify gaps in the system, and update responses as institutional policies evolve. By centralizing the information in this structured manner, the system achieves low computational overhead while maintaining high response accuracy.

Knowledge-based systems are particularly suitable for educational and administrative chatbots, where the priority is factual accuracy and reliability rather than predictive learning from unstructured data. Unlike machine learning-driven chatbots, which require extensive training datasets and continuous model updates, a rule- and knowledge-based system provides deterministic and verifiable responses. This ensures that users—students, staff, or visitors—receive consistent and authoritative information.

4.7 System Implementation

This section describes the practical development of the web-based informational chatbot, detailing both the frontend and backend components. Implementation involves translating the design models, knowledge base, and system architecture into a functional system that meets the needs of students and staff at the School of Computing, Federal Polytechnic Ede.

4.7.1 Frontend Implementation

The frontend of the chatbot system is designed using standard web technologies, including HTML5, CSS3, and JavaScript, to ensure broad compatibility across modern browsers and devices. The interface provides a message display area where both user inputs and chatbot responses are shown in a conversational style, creating a familiar chat environment. An input

field allows users to type their questions naturally, while a "send" button or enter key triggers message submission.

To enhance responsiveness and user experience, the frontend employs asynchronous communication techniques (AJAX), enabling the system to send and receive messages from the server in real-time without requiring a page reload. This design ensures smooth interaction, reduces latency, and gives the impression of a live conversation.

The user interface follows human-centered design principles, prioritizing clarity, accessibility, and ease of use. Visual elements, such as contrasting colors for messages, scrollable conversation history, and notification indicators, improve usability and help users interact with the chatbot efficiently. By incorporating these design considerations, the system caters to users of varying technical proficiency while encouraging frequent engagement (ISO 9241-210, 2019).

4.7.2 Backend Implementation

The backend of the system handles all processing, decision-making, and data management. Built using a server-side scripting language such as PHP, it serves as the core engine that interprets user queries, interacts with the knowledge base, and generates appropriate responses.

When a user submits a query, the backend first performs input validation and sanitization to prevent security vulnerabilities such as SQL injection or cross-site scripting. The system then preprocesses the query by normalizing text (e.g., converting to lowercase, removing unnecessary symbols) to improve keyword or intent matching.

The knowledge base is queried for entries that correspond to the identified intent or keywords. If a matching entry is found, the corresponding response is retrieved and returned

to the frontend. If no exact match exists, the system generates a fallback response that prompts the user to rephrase the question or consult an administrator.

In addition to processing queries, the backend records interaction logs, which include user queries, timestamps, and response details. These logs are essential for monitoring system usage, analyzing frequently asked questions, and guiding knowledge base updates to improve chatbot accuracy over time.

This backend implementation aligns with modern web-based chatbot architectures, combining efficiency, reliability, and maintainability. By separating processing logic from presentation, the system ensures that updates to the knowledge base or server-side scripts do not disrupt the user interface, thereby supporting scalability and long-term sustainability (Shawar & Atwell, 2021).

4.8 Security Considerations

Security is an important aspect of the system design. User inputs are sanitized to prevent common web vulnerabilities such as SQL injection and cross-site scripting. Administrative access is restricted to authorized personnel to protect the integrity of the knowledge base. These measures are consistent with web application security best practices outlined by OWASP (2023).

4.9 Summary

This chapter discussed the design and implementation of the web-based informational chatbot developed for the School of Computing, Federal Polytechnic Ede. It covered the system architecture, requirements, design models, chatbot logic, implementation strategies, and security considerations. The next chapter will focus on system testing, evaluation, and performance analysis.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary

This project addressed the design and implementation of a web-based informational chatbot for the School of Computing, Federal Polytechnic Ede, with the aim of improving access to academic and administrative information. The motivation for the study stemmed from the observed challenges faced by students and staff in obtaining timely and consistent information through traditional enquiry channels, which are often manual, time-bound, and dependent on the availability of administrative personnel.

The study began with a comprehensive review of related literature on chatbot systems, conversational agents, and web-based information systems. These studies provided insight into different chatbot design approaches, particularly rule-based, knowledge-based, and machine learning-driven models. Based on the nature of institutional information and the need for accuracy and control, a knowledge-based, rule-driven approach was adopted for the system.

System analysis and design were carried out to define functional requirements, system architecture, and interaction flow. The system was structured using a three-tier architecture comprising the presentation layer, application layer, and knowledge base. Implementation was achieved using standard web technologies, ensuring compatibility, ease of deployment, and maintainability. The chatbot was designed to accept user queries, process them using predefined rules and keyword mappings, and return appropriate responses in real time.

The results of the implementation demonstrate that the system effectively supports information dissemination, reduces repetitive enquiries, and provides a convenient platform

for users to interact with institutional information resources. The project therefore confirms the relevance and applicability of chatbot technology within academic environments.

5.2 Conclusion

The successful implementation of the web-based informational chatbot shows that conversational systems can play a significant role in enhancing administrative efficiency and user satisfaction in tertiary institutions. By providing instant access to frequently requested information, the chatbot reduces dependence on manual enquiry processes and minimizes delays associated with human-mediated communication.

The system met its stated objectives by delivering a functional, user-friendly, and reliable chatbot that operates within the technological constraints of the institution. The adoption of a knowledge-based design ensured that responses remained consistent, verifiable, and easy to update, which is particularly important in an academic setting where information accuracy is critical.

Furthermore, the project demonstrates that sophisticated machine learning techniques are not always necessary for effective chatbot deployment, especially in domains where information is structured and predictable. The findings reinforce existing research that supports the continued relevance of rule-based and knowledge-driven chatbots for informational and administrative purposes (Russell & Norvig, 2021). Overall, the developed system contributes positively to digital transformation efforts within the School of Computing, Federal Polytechnic Ede, and provides a foundation upon which more advanced conversational systems can be built.

5.3 Recommendations

Based on the findings and experience gained during the development of this system, several recommendations are made. First, it is recommended that the chatbot's knowledge base be continuously maintained and updated to reflect changes in academic calendars, course offerings, departmental policies, and administrative procedures. Regular updates will ensure sustained accuracy and relevance of the system.

Secondly, future enhancements may involve the integration of natural language processing (NLP) techniques to improve the chatbot's ability to understand more complex sentence structures, synonyms, and ambiguous queries. This would increase flexibility and user satisfaction while complementing the existing rule-based framework.

In addition, the system can be improved by incorporating user feedback and analytics features, which would allow administrators to monitor system performance, identify frequently asked questions, and refine responses based on actual usage patterns. Such data-driven improvements would enhance the overall effectiveness of the chatbot.

sssIt is also recommended that the chatbot be fully integrated into the institution's official website or student portal, making it easily accessible to all users. Extending access to mobile platforms and supporting multiple languages could further broaden usability and inclusiveness.

Finally, future research may explore the application of intelligent chatbots across other departments within the institution or compare the performance of rule-based and machine learning-based chatbot models in similar academic environments. This would contribute to a deeper understanding of conversational systems and their role in educational service delivery (Shawar & Atwell, 2021).